

Project Report

B105 Applied Statistical Modelling

Title

Statistical Analysis of Restaurant Tips Amount and Customer Satisfaction Using the tipsn.csv Dataset

Introduction

Restaurant keep track of the customers, how many customers are coming to visit restaurant.

What is pattern of spending of them. How much they spend on tips. What size or group in which they generally comes.

In this statistical analysis I have used tipsn dataset. The main aim is to identify number of customers visiting , How many customers what is gender proportion of them. In What groups they are coming like group of 1,2,3,4, etc. other patterns like smoking , days effect on tips or bills. All monetary values used in the project are in **euros (€)**.

The statistical analysis includes following important points to be covered

Like data preparation, Cleaning of the data, use of sampling, EDA, descriptive analysis, testing of the hypothesis, analysis of the correlation, analysis of the regression, Confidence interval checking, Finding and suggesting Business Recommendations, Limitations of the analysis, and future work that has to be performed

Business Problem

The restaurant is looking for the studying the pattern of tip received from the customers. What is the impact on tips and bills of day, time, size of the customers.

This helps to understand the customer behavior and ideas to improve customer satisfaction and to increase the revenue.

Business Questions

Few of the questions we are looking for answer based on analysis includes

Question 1

Is the total bill is effected by the tip amount given by each customer?

Question 2

Is any difference in lunch and dinner average tips values in Euro?

```
#import the required dataset from system
tips <- read.csv("tipsn.csv")
# Display first few rows from the dataset
head(tips)
# Display dimensions
dim(tips)
# Display summary
summary(tips)
```

Question 3

Is any difference in the average tips that were given by the customers gender wise?

Question 4

Is size of the group

effect the average tip amount like if more number of people in group they are generally pay higher tip.

Question 5

Is the tip amount effected by total bill, table size, characteristics of the customers

Dataset Description

The uploaded dataset which is used for testing purpose is

244 observations and 7 features or variables

The variables includes are

- **Total_bill** → it is of **numerical type** , and it is total bill amount paid by the customer in €
- **Tip** → it is **numerical type data**, and it is tip amount paid by the customer in € it is an important amount which is used to find satisfaction of the customer visiting the restaurant.
- **Smoker** → it is of **categorical type data** which can be yes or no
- **Day** → it is of **categorical type data** which can be some working day in restaurant when customer visits the restaurant for lunch or Dinner
- **Time** → it is of **categorical type data** which can be Lunch or Dinner
- **Size** → it is **numeric data but can be used in category as size may vary from 1 to 6**
And can be divided in 6 categories or can be used as numeric

Code of Summary of the Dataset

Step 1: Loading Dataset and displaying records and size of the dataset Code window

```
1 #import the required dataset |from system
2 tips <- read.csv("tipsn.csv")
3 # Display first few rows from the dataset
4 head(tips)
5 # Display dimensions
6 dim(tips)
7 # Display summary
8 summary(tips)
```

Output Screen1

```
> head(tips)
  total_bill tip gender smoker day  time size
1    16.99 1.01 Female    No  Sun Dinner   2
2    10.34 1.66  Male    No  Sun Dinner   3
3    21.01 3.50  Male    No  Sun Dinner   3
4    23.68 3.31  Male    No  Sun Dinner   2
5    24.59 3.61 Female    No  Sun Dinner   4
6    25.29 4.71  Male    No  Sun Dinner   4
```

Output Screen 2

```
> dim(tips)
[1] 244  7
> summary(tips)
  total_bill      tip      gender      smoker      day      time      size
Min.   : 3.07   Min.   : 1.000   Female: 87    No :151   Fri :19   Dinner:176   Min.   :1.00
1st Qu.:13.35   1st Qu.: 2.000   Male  :157   Yes: 93   Sat :87   Lunch : 68    1st Qu.:2.00
Median :17.80   Median : 2.900                                Sun :76                                Median :2.00
Mean   :19.79                                Thur:62                                Mean   :2.57
3rd Qu.:24.13   3rd Qu.: 3.562                                3rd Qu.:3.00
Max.   :50.81   Max.   :10.000                                Max.   :6.00
```

Step 2: Checking the missing values in the dataset, as the dataset is clean so we find no values are missing in the dataset.

```
> colSums(is.na(tips))
total_bill      tip      gender      smoker      day      time      size
         0         0         0         0         0         0         0
```

Step 3

Checking the duplicates records in the dataset, may be some false data or synthetic data in the dataset

```
> sum(duplicated(tips))  
[1] 1
```

As this shows that the dataset has exactly 1 duplicate data row

```
> tipsnew<- tips[!duplicated(tips), ]  
> dim(tipsnew)  
[1] 243 7  
> |
```

1 duplicate row is deleted

Step 4: converting categorical to factors

```
#converting categorical to factors
tipsnew$gender <- as.factor(tipsnew$gender)
tipsnew$smoker <- as.factor(tipsnew$smoker)
tipsnew$day <- as.factor(tipsnew$day)
tipsnew$time <- as.factor(tipsnew$time)
str(tipsnew)
```

output

```
'data.frame': 243 obs. of 7 variables:
 $ total_bill: num 17 10.3 21 23.7 24.6 ...
 $ tip : num 1.01 1.66 3.5 3.31 3.61 4.71 2 3.12 1.96 3.23 ...
 $ gender : Factor w/ 2 levels "Female","Male": 1 2 2 2 1 2 2 2 2 2 ...
 $ smoker : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ day : Factor w/ 4 levels "Fri","Sat","Sun",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ time : Factor w/ 2 levels "Dinner","Lunch": 1 1 1 1 1 1 1 1 1 1 ...
 $ size : int 2 3 3 2 4 4 2 4 2 2 ...
```

Sampling

The original dataset had 244 records and after removing the duplicates I have 243 instances of all the records to avoid any type of loss of information. Generally sampling is done when dataset is generally very large. But for just finding and demonstrate purpose we are taking 85% of the dataset sampling but work on actual dataset of 243 only as data set is not very large.

Sampling code is optional in this analysis

```
#sampling for illustration
set.seed(111)
sampldata<- tipsnew[ sample(
  1:nrow(tipsnew),
  size = round(0.85*nrow(tipsnew))
),
]
dim(sampldata)
```

Output:

```
> set.seed(111)
> sampldata<- tipsnew[ sample(
+   1:nrow(tipsnew),
+   size = round(0.85*nrow(tipsnew))
+ ),
+ ]
> dim(sampldata)
[1] 207  7
```

Descriptive statistics

Descriptive statistics is performed on some important columns which is used to find the average total bill, median or central value of the total bill as well as to find out the spread of the total bill using standard deviation, also the minimum and maximum amount is calculated on total bill, tip, tablesize

```
#Descriptive statistics
mean(tipsnew$total_bill)
median(tipsnew$total_bill)
sd(tipsnew$total_bill)
min(tipsnew$total_bill)
max(tipsnew$total_bill)
quantile(tipsnew$total_bill)
```

```
> mean(tipsnew$total_bill)
[1] 19.81387
> median(tipsnew$total_bill)
[1] 17.81
> sd(tipsnew$total_bill)
[1] 8.910071
> min(tipsnew$total_bill)
[1] 3.07
> max(tipsnew$total_bill)
[1] 50.81
> quantile(tipsnew$total_bill)
 0%   25%   50%   75%  100%
3.070 13.380 17.810 24.175 50.810
>
```

The average restaurant bill is calculated as approximately **€19.79**

The median showing is that half of the customers are paying below **€17.80** and half of the customers paying above **€17.80**

The maximum bill is **€50.81**, the minimum bill found is **€3.07**.

We can see the big gap between the minimum and maximum value of the bill it shows that customer spending requirements varied.

Code for descriptive statistics of tip

```
#Descriptive statistics tip
mean(tipsnew$tip)
median(tipsnew$tip)
sd(tipsnew$tip)
min(tipsnew$tip)
max(tipsnew$tip)
quantile(tipsnew$tip)
```

```
7 | (Top Level) -
```

```

> mean(tipsnew$tip)
[1] 3.002387
> median(tipsnew$tip)
[1] 2.92
> sd(tipsnew$tip)
[1] 1.385002
> min(tipsnew$tip)
[1] 1
> max(tipsnew$tip)
[1] 10
> quantile(tipsnew$tip)
 0%   25%   50%   75%  100%
1.000 2.000 2.920 3.575 10.000

```

Tips pattern shows that minimum tip amount is 1 euro, while the maximum tip amount is 10 euro on an average tip amount is near to 3 euro, half of the customers are paying tip below 2.92 euro and rest half are paying above 2.92 euro.

Code for descriptive statistics of size

```

#Descriptive statistics size
mean(tipsnew$size)
median(tipsnew$size)
sd(tipsnew$size)
min(tipsnew$size)
max(tipsnew$size)

```

Output:

```

> max(tipsnew$size)
[1] 6
> mean(tipsnew$size)
[1] 2.572016
> median(tipsnew$size)
[1] 2
> sd(tipsnew$size)
[1] 0.9523562
> min(tipsnew$size)
[1] 1
> max(tipsnew$size)
[1] 6
> quantile(tipsnew$size)
 0%  25%  50%  75% 100%
 1   2   2   3   6
>

```

Size pattern shows that most of the people visited in restaurant in group of 2 more than 50% of people are in group of 2 and 75% of the customers are in the group of 3 or less, so seating arrangement such that should be in restaurant so that 2, 3 people can easily accommodate and maximum number of orders can be entertained.

```
> table(tipsnew$gender)
```

Female	Male
86	157

As per the analysis more male customers are visiting restaurant as 157 Male , 86 female customers are visible in the dataset.

```
> table(tipsnew$smoker)
```

No	Yes
151	92

According to smoker count 151 are non smokers and 92 are smokers

```
> table(tipsnew$time)
```

Dinner	Lunch
176	67

According to the given dataset 176 are in dinner, 67 in lunch so more number of people are visited in Dinner so restaurant should have more arrangement in Dinner time instead of Lunch as large number of customers are visiting in Dinner time only.

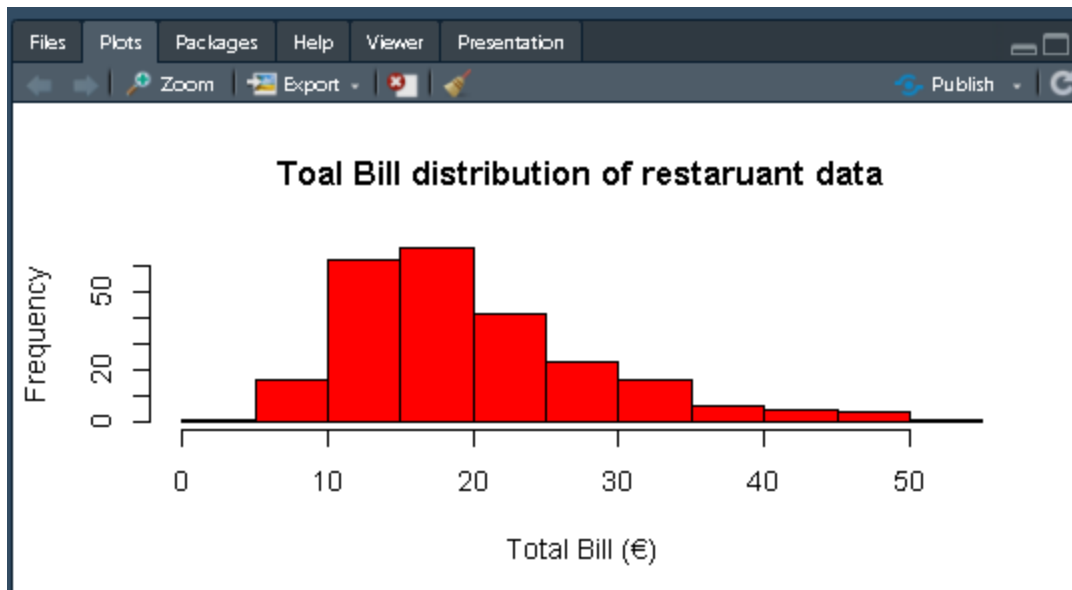
```
> table(tipsnew$day)
```

Fri	Sat	Sun	Thur
19	87	76	61

According to days wise analysis most of the customers are visiting in 3 days Saturday, Sunday, Thursday and very less number of customers on Friday and no customers on Monday, Tuesday, Wednesday as per the data collected

Total Bill Distribution

```
hist(  
  tipsnew$total_bill,  
  main = "Total Bill distribution of restaurant data",  
  xlab = "Total Bill (€)",  
  ylab = "Frequency",  
  col = "red",  
  border = "black"  
)
```

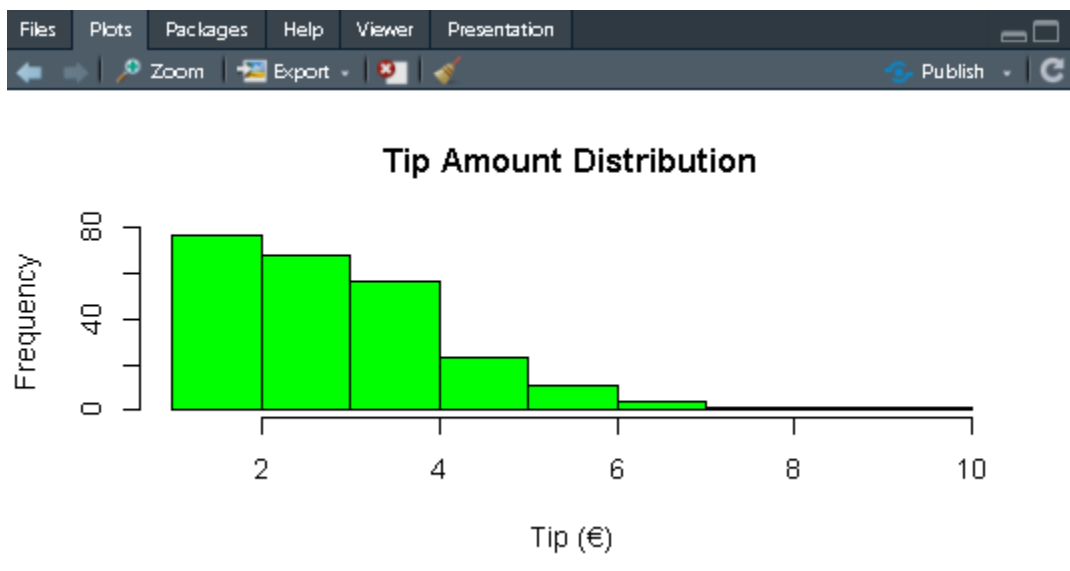


Most of the bills is towards 30Euro or lower it is towards left side of the histogram.

Tip Distribution of the data

```
hist(
  tipsnew$tip,
  main = "Tip Amount Distribution",
  xlab = "Tip (€)",
  ylab = "Frequency",
  col = "green",
  border = "black"
)
```

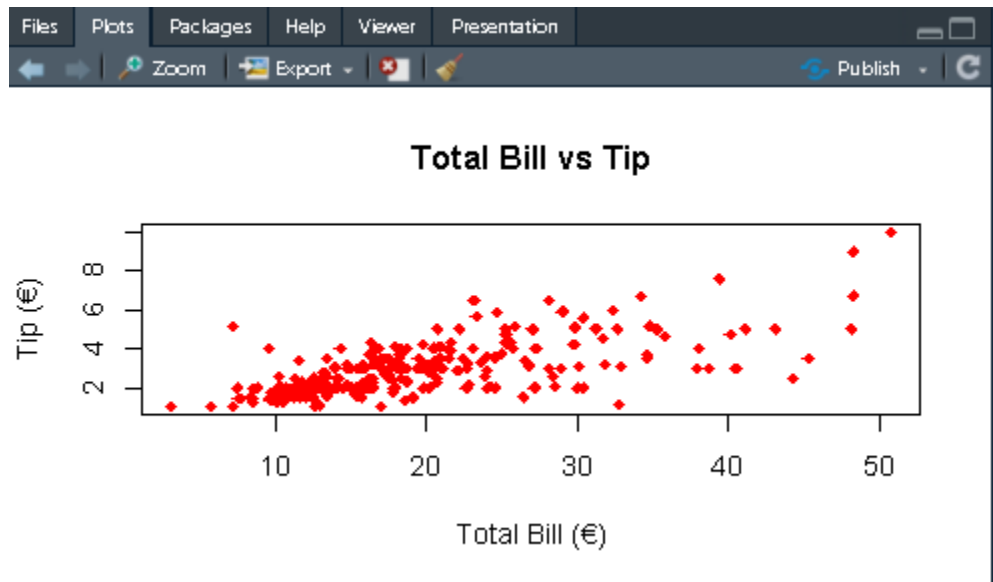
Output of the code



Most of the tips are 6 euro or less it is maximum between 1 to 3 Euro which looks realistic

```
plot(
  tipsnew$total_bill,tipsnew$tip,main="Total Bill vs Tip",col='red',xlab="Total Bill (€)",ylab="Tip (€)",pch=18)
,
```

OUTPUT:



Graph shown an upward trend which means with increase in total bill customers are paying more tip amount

Hypothesis:

Null hypothesis : H_0

There is no relationship between tip amount and total bill

Alternative hypothesis — H_1

As shows there is linear relationship between tip amount and the total bill as total bill increases tip amount is increasing

Significance level

We are considering

$\alpha = 0.05$

```
cor.test( tipsnew$total_bill, tipsnew$tip, method = "pearson" )
```

```
Pearson's product-moment correlation

data: tipsnew$total_bill and tipsnew$tip
t = 14.202, df = 241, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.6001314 0.7381411
sample estimates:
      cor 
0.6749979
```

For the dataset after cleaning

$r \approx 0.675$

$p < 0.001$

Interpretation

Correlation between tip amount and total amount is 0.675, which indicates strong positive relationship between these two. P value observed is very small much smaller than 0.05.

Therefore we can reject H_0

There is a strong evidence of positive relationship hence with the increase in total bill at restaurant tip amount is also increasing.

This is an association not a proof that for every higher bill there is higher amount of tip this is just a trend

Second Hypothesis

H_0 (Null Hypothesis)

The average of tip amount for lunch and dinner are same

H_1 (alternative Hypothesis)

The average of tip amount for lunch and dinner are differ with each other

Selecting the Test

As per in the given dataset Tip column is a numerical variable and Lunch and Dinner is in time column and has Categorical data having two independent groups Lunch or Dinner.

We are using Welch's t-test which is used to consider generally the variance which are not same

Coding

```
tm_test<-t.test(tip ~ time, data = tipsnew, var.equal = FALSE)
tm_test
|
```

output

```
Welch Two Sample t-test

data: tip by time
t = 1.9839, df = 140.41, p-value = 0.04922
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.001261242 0.726169220
sample estimates:
mean in group Dinner mean in group Lunch
      3.102670         2.738955
```

Interpretation

For the dataset after cleaning

- $t \approx 1.984$

- $p \approx 0.0492$

as visible from the p value

$$0.0492 < 0.05$$

We reject the H_0 Null Hypothesis at level of 5%

There is a clear statistical difference between the tips of lunch and dinner.

The average dinner tip is calculated as nearly **€3.10**, as compared with the lunch tip which nearly y **€2.74** for lunch.

What important we can observe that

Also we can also see that p-value is very near to 0.05.

This result is borderline at the chosen 5% level not a very strong evidence as p value is very near to 0.05

Hypothesis 3

H_0 (Null hypothesis)

There is no difference between female and male customers for the average tip amount

H_1 (alternative hypothesis)

There is difference between female and male customers for the average tip amount

Selecting the Test

As per in the given dataset Tip column is a numerical variable and gender column value has male and female Categorical data having two independent groups male and female as per the dataset

We are using Welch's t-test which is used to consider generally the variance which are not same

```
gender_test<- t.test( tip ~ gender,  data = tipsnew,  var.equal = FALSE)
gender_test
```

Output

```
Welch Two Sample t-test

data:  tip by gender
t = -1.4267, df = 212.77, p-value = 0.1551
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.5870094  0.0940528
sample estimates:
mean in group Female  mean in group Male
          2.843140          3.089618
```

Interpretation

As per the finding of the test

0.155 > 0.05

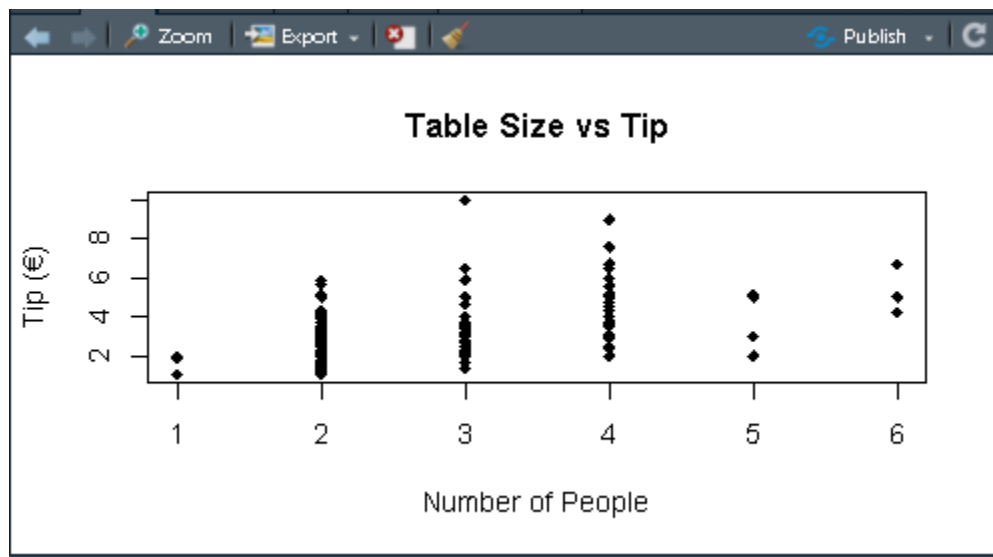
We can not reject the null hypothesis

So the given dataset not provides sufficient statistical information in the average tip difference between male and female customers though the female group tip amount is 2.843 which is not much differing from male customers having average tip amount 3.0896

Although the sample average seen is slightly higher for male customers but the difference is not statistically significant.

Checking impact of size of the group on tip amount to see a trend whether the size of the group effecting tip amount

Using scatter plot to see the general trend



```
> cor.test( tipsnew$size, tipsnew$tip, method = "pearson")
```

Pearson's product-moment correlation

data: tipsnew\$size and tipsnew\$tip

t = 8.6888, df = 241, p-value = 5.688e-16

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.3862989 0.5786778

sample estimates:

cor

0.4884004

Result for interpretation

$r \approx 0.488$

$p < 0.001$

Interpretation

There is moderate relationship visible which is on positive side with the size of the group total tip amount increases. Larger the groups more chances of average total top

Mutiple regression analysis based on total_bill, size of the group, gender, smoking status, visiting time.

```
multiple_model<-lm( tip ~ total_bill + size + gender + smoker + time, data = tipsnew )
summary(multiple_model)
```

Output

```
Call:
lm(formula = tip ~ total_bill + size + gender + smoker + time,
    data = tipsnew)

Residuals:
    Min       1Q   Median       3Q      Max
-2.8923 -0.5787 -0.0563  0.4971  4.0671

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.726357   0.232948   3.118  0.00205 **
total_bill   0.094006   0.009527   9.867 < 2e-16 ***
size         0.180443   0.088345   2.042  0.04221 *
genderMale  -0.030027   0.140609  -0.214  0.83108
smokerYes   -0.081179   0.139533  -0.582  0.56126
timeLunch   -0.002068   0.151580  -0.014  0.98913
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.021 on 237 degrees of freedom
Multiple R-squared:  0.4677,    Adjusted R-squared:  0.4565
F-statistic: 41.65 on 5 and 237 DF,  p-value: < 2.2e-16
```

The dataset gives nearly $R^2 = 0.468$, Adjusted $R^2 = 0.457$

Interpretation of Multiple Regression

Total bill is an important factor for the tip amount

As we can see the The coefficient is nearly **0.094**

So, we can see a €1 increase in bill is associated with approximately **€0.094 higher tip**.

Table size or number of group in size is another important factor for considering the amount on tip

Calculated coefficient is **0.180**

This gives an idea that after an additional person at a table tip amount is about **€0.18 higher total tip**, on average.

The p-value calculated also is approximately **0.042**, which is below 0.05.

Gender , the gender coefficient is not much important impact statistically

$p \approx 0.831$

Therefore, there is not a very strong statistical evidence based on gender effect in this model.

Smoker status, The smoker calculated coefficient is **$p \approx 0.561$** so there is not much significant impact on tip value based on smoker status.

Hypothesis	Test	Result	Decision
Bill and tip have no relation	Pearson correlation	$p < 0.001$	Reject H_0
Lunch & dinner have equal amount of value	Welch t-test	$p \approx 0.049$	Reject H_0 at 5%
Male and female have equal avg tip amount	Welch t-test	$p \approx 0.155$	Do not reject H_0
Size has no impact on average tip amount	Pearson correlation	$p < 0.001$	Reject H_0

Data-Driven Recommendations

Some important recommendations based on all statistical analysis

Recommendation 1 :

Bill amount is an important factor for calculating the tip amount as it has strong impact on tip amount.

Since total bill and tip have a strong positive relationship, the restaurant can use expected bill amounts to estimate expected tip revenue.

Recommendation 2 :

Considering table size or size of group

Larger the group higher will be the chances of total tip and total amount

Size can be considered as forecasting total bill or tip revenue and the amount increases for total bill

and tip with the increase in

Recommendation 3 :

Gender is not an important factor for predicting or trend for total amount or tip amount, analysis based on statistics does not provide significant evidence for trend that supports gender vs bill
So gender should not be considered as important factor for analysis.

Recommendation 4 :

Though dinner has a higher average tip in the basic comparison to the Lunch.
Meal time is not much significant on overall bill amount or tip amount
Therefore, the restaurant should not assume that dinner only required to have higher tips.

Recommendation 5:

We can collect more better customer service information, few of the other factors can also be considered for better understanding of the tips like service rating, waiting time, food rating, payment method, offers and discount, restaurant location etc for better analysis of the customer satisfaction

Limitations

Few of the limitations that are visible in the project that can be worked upon and can be improved in future

- the data set that I have used for statistically in the current state is smaller that can be in future to have larger dataset for better results and analysis.
- Features and variables that are available in the dataset is small in size we have 7 features to keep information about the restaurant
- The regression are not properly normally,distributed in the given dataset.
- Variance in the residual is not perfectly found to be constant.
- More advanced statistical approach can be used in future for future prediction
- Results can not be applied to all restaurants or on all customers the dataset is small so not covers wider range of hotel industry in current analysis.

Future Work

Future work can include larger dataset that should cover different restaurants from different cities, also data can be collected for different months, different seasons on large number of customers

Additional variables can be keep track of like service rating, food rating, waiting time in the restaurant, discounts,offers etc.

Advance models can be used in future for better analysis like multiple linear regression ,Random Forest , Gradient Boosting, XGBoost , KNN, Ensemble etc. Cross validation can be used to check how well model can work on unseen data.

Final Conclusion

Conclusion:

Statistical analysis is based on dataset of a restaurant which is available in the csv file tipsn.csv where all monetary or money related values are in the euros. After finding and analysis in preprocessing stage data set is checked for missing values or duplicate values and that was cleaned before any further statistical analysis. We have 243 total observations these all are real no fabricated or synthetic data is added. Average bill found to be near about €19.79 and the average tip was near about €3.00. as per the statistical data analysis significantly positive relationship between total bill and tip amount as with the increase in total bill clear evidence of increase in tip amount. There was also evidence of a difference between lunch and dinner tips at the 5% significance level, while the difference between male and female customers was not much statistically significant. size also showed a positive relationship with tip amount as with the size increases tip amount increases.

Most of the groups are smaller which are generally 2,3,4 so more priority should be given facilities based on these groups. The regression analysis showed that total bill has impact on the approximately 45.6% of the changes in the tips, while the multiple regression model explained and covered various parameters that are effecting the tips amount which is approximately 46.8%. Overall, bill amount as well as table size appear to be useful variables for understanding restaurant tip revenue and total revenue as tip, total amount and size all are significantly correlated with each other. Also weekends in the dinner shows more number of people visiting so proper arrangement should be provided by the hotel to handle increased customer flow in weekend with good quality services. However, the dataset does not include some of the important factors such as service quality and customer satisfaction, so future analysis using a larger and more detailed dataset could provide better results and statistical results.